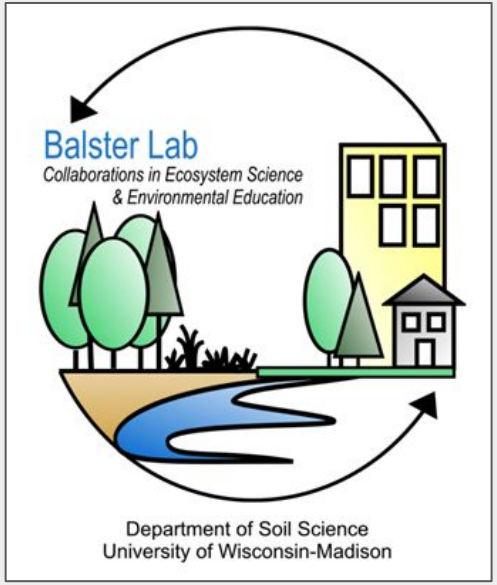
Determining the Correlation between Soil pH, Soil Volumetric Moisture Content, and the Frequency of Damping off Disease through Fusarium spp in a Controlled Greenhouse Setting

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Wilson State Nursery, Boscobel WI

- Established 1951
- Managed by the WDNR
- Important for reforestation in WI
    - 1.1 million red pine seedlings in 2022
- Vital to local economy and community



(WSN Management, 2022)



Damping Off Disease

- Fungal pathogen-caused disease that infects trees (in this case)
- Infects their roots and base of the stem
- Only preventable, not curable
- Infects before 6 week mark (for red pine)
  - Before woody tissue develops
  - Lamichhane et al., 2017
- Common preventative measures:
  - Fungicides
  - Drive down pH (Cruz et al., 2020)
  - Cleaning equipment
  - No excess moisture



Example of healthy red pine seedlings in the “bird cage” stage



Example of a red pine seedling dying from damping off disease



Patches of dead seedlings from damping off

Seedlings that likely survived the disease, but are permanently stunted; not ideal for selling the trees later on

Hypothesis

We hypothesize that the severity of damping off, measured by frequency of death, of response to Fusarium by red pine seedlings will increase with soil moisture content and decrease with lower soil pH.

Methods

- **Barrel setup**
  - 14 total barrels planted – 2 barrels not enough germinated seedlings
  - 2 ft of WSN soil (sandy loam)
  - Each half of each barrel had 1 sensor, 2 inches below surface
    - 5-TM METER Temperature and Moisture Sensor probe
    - 1 sensor placed lower in barrel (~12 inches from surface)
- **4 Treatment groups**
  - High moisture/ambient pH
  - Low moisture/ambient pH
  - High moisture/lower pH
  - Low moisture/lower pH
  - Each barrel was assigned either low/high moisture, and ambient/low pH
- **Barrel soil pH**
  - Soil pH was adjusted with 200mL solution of 1.3 g ammonium sulfate and DI water
  - Tested every 3 days and adjusted accordingly
- **Barrel soil moisture**
  - 7/14 barrels chosen as high moisture, other half were low moisture
  - High moisture: 0.12-0.14 m^3 water/ m^3 soil
  - Low moisture: 0.09-0.11 m^3 water/ m^3 soil

Methods

- **Sewing red pine seeds in barrels**
  - Each side of each barrel sewn with 20 seeds
  - 1 cm depth using sterile tweezers
  - Seeds not sterilized before planting
- **Fusarium inoculant creation**
  - Culture grown in WDNR lab, collected by plant pathologist from WSN
  - Once plate full of colonies–
    - Add 10 mL DI water and 0.05% v/v Tween 20
    - Mycelia and spores scraped off
    - Mixed filtered through four sterile layers of cheesecloth
    - Inoculum diluted to a fairly strong strength (1e6 spores/mL)
    - Kept at 10C for up to one week
- **Inoculation of red pine seedlings**
  - Each barrel inoculated when at least 7 out of 20 seedlings emerged
  - Sprayed inoculum on soil and seedlings until runoff
- **Monitoring seedlings**
  - Seedlings were declared dead after 3 days with clear damping off symptoms
  - All seedlings per barrel were harvested exactly 31 days after inoculation (well-after birdcage stage)

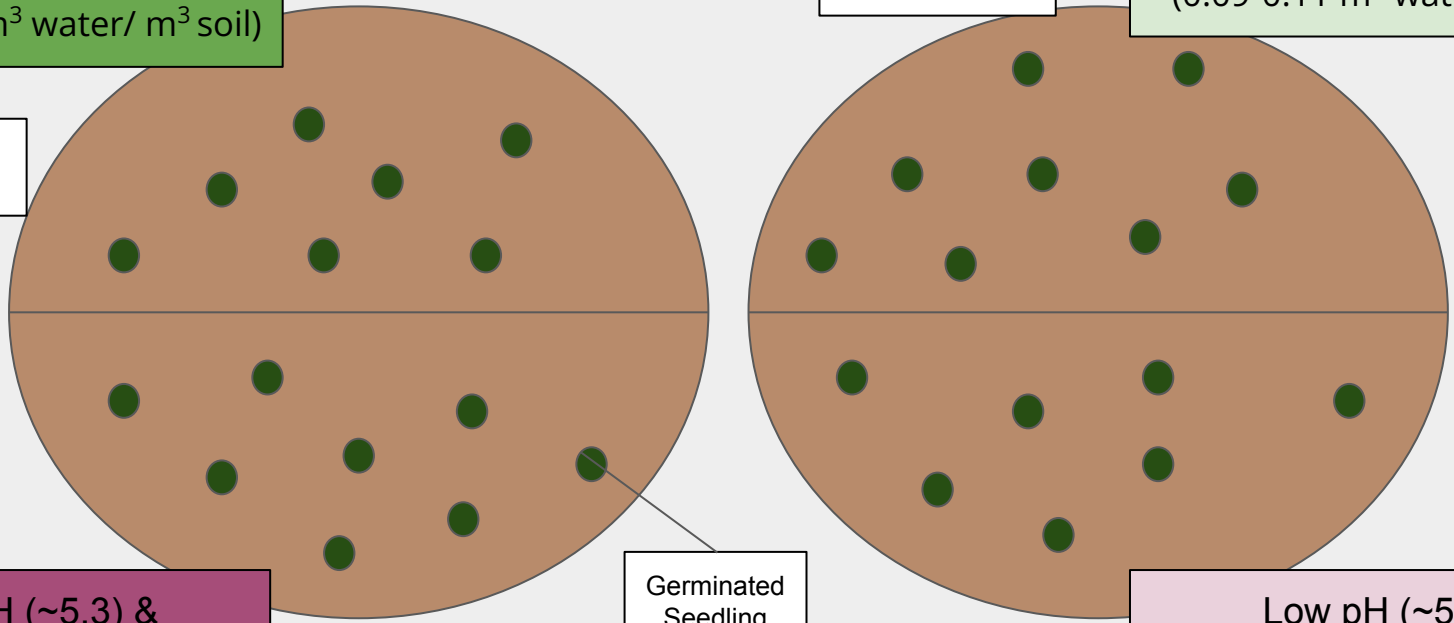
Methods

Ambient pH (~7) &
High Moisture
(0.12-0.14 m^3 water/ m^3 soil)

Example Barrel 1:
High Moisture

Example Barrel 2:
Low Moisture

Ambient pH (~7) &
Low Moisture
(0.09-0.11 m^3 water/ m^3 soil)



Germinated
Seedling

Low pH (~5.3) &
High Moisture
(0.12-0.14 m^3 water/ m^3 soil)

Low pH (~5.3) &
Low Moisture
(0.09-0.11 m^3 water/ m^3 soil)

Methods





Methods



Treatment Group	Sample Size	# Seedlings Dead
AP/HM	65	12
AP/LM	44	16
LP/HM	49	5
LP/LM	47	11
	205 total	44

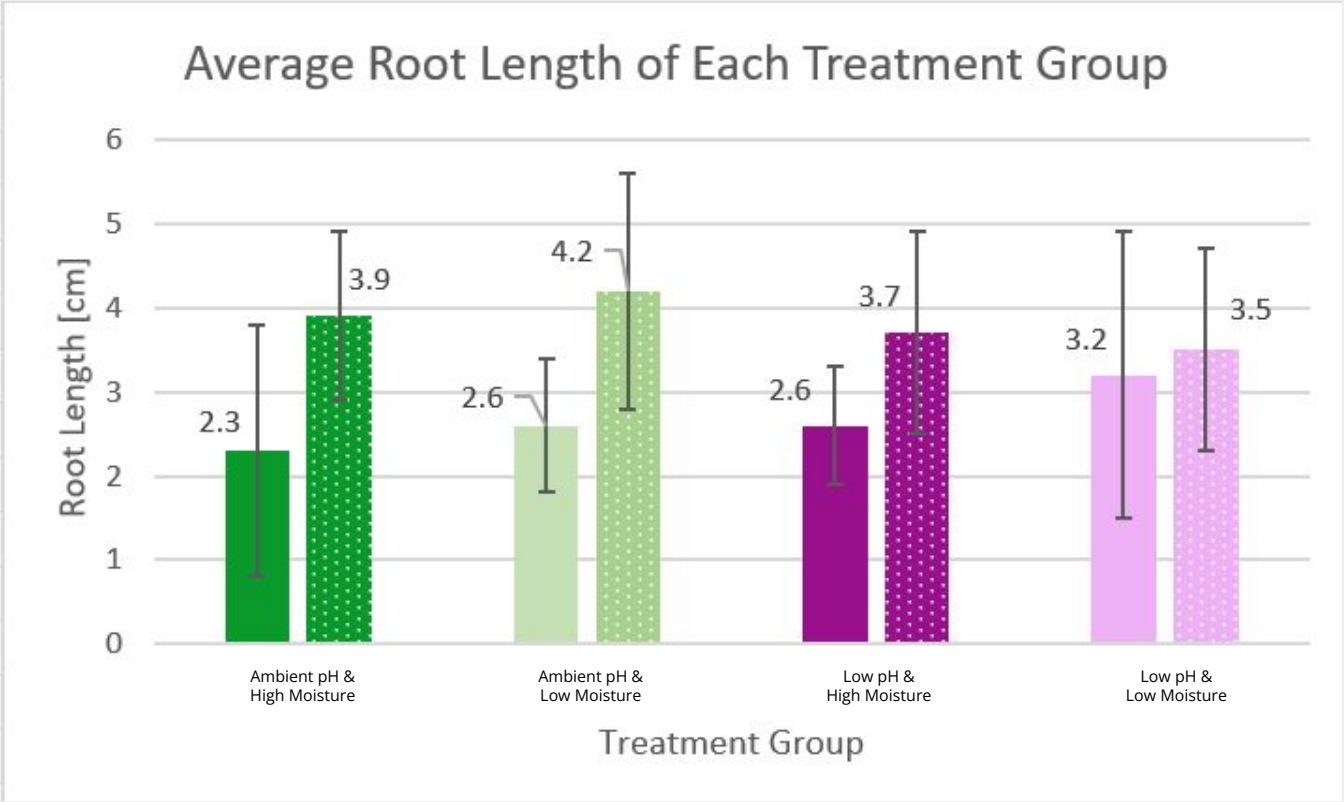
Results

- Damping Off Frequency:
  - AP/LM: highest seedling death frequency
  - LP/HM: lowest seedling death frequency
- Days Between Inoculation and Death
  - LP/LM: longest average time
  - AP/LM: shortest average time
- **AP/LM killed the most seedlings in the shortest amount of time**

	Ambient pH + High moisture	Ambient pH + Low moisture	Low pH + High moisture	Low pH + Low moisture
Average Time Between Inoculation and Death *[Days]*	18 sd: 7	15 sd: 3	20 sd: 13	22 sd: 6
Total Deaths from Damping Off *[Number of Seedlings]*	12	16	5	11
Percent of Seedlings Dead from D.O. *(# of seedlings dead per category/ total # of seedlings per category)*	18.5 %	36.4%	10.2%	23.4%

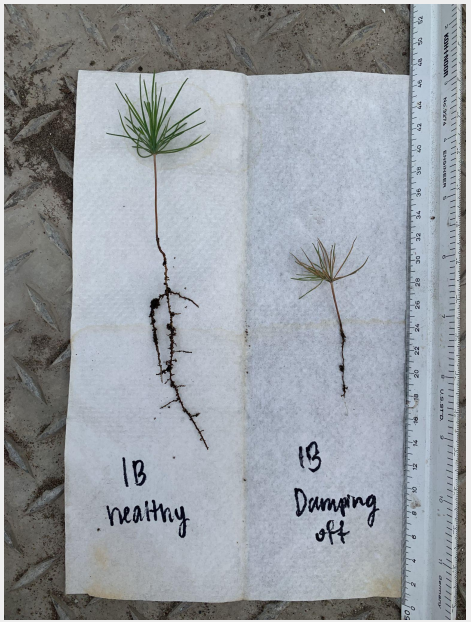
Table 1: Average time in days between inoculation and seedling death (row 2) and the total deaths from damping off in number of seedlings per treatment group (row 1). The percentages in row 2 represent the percent of seedlings that died with clear damping off symptoms in that specific category.

Results

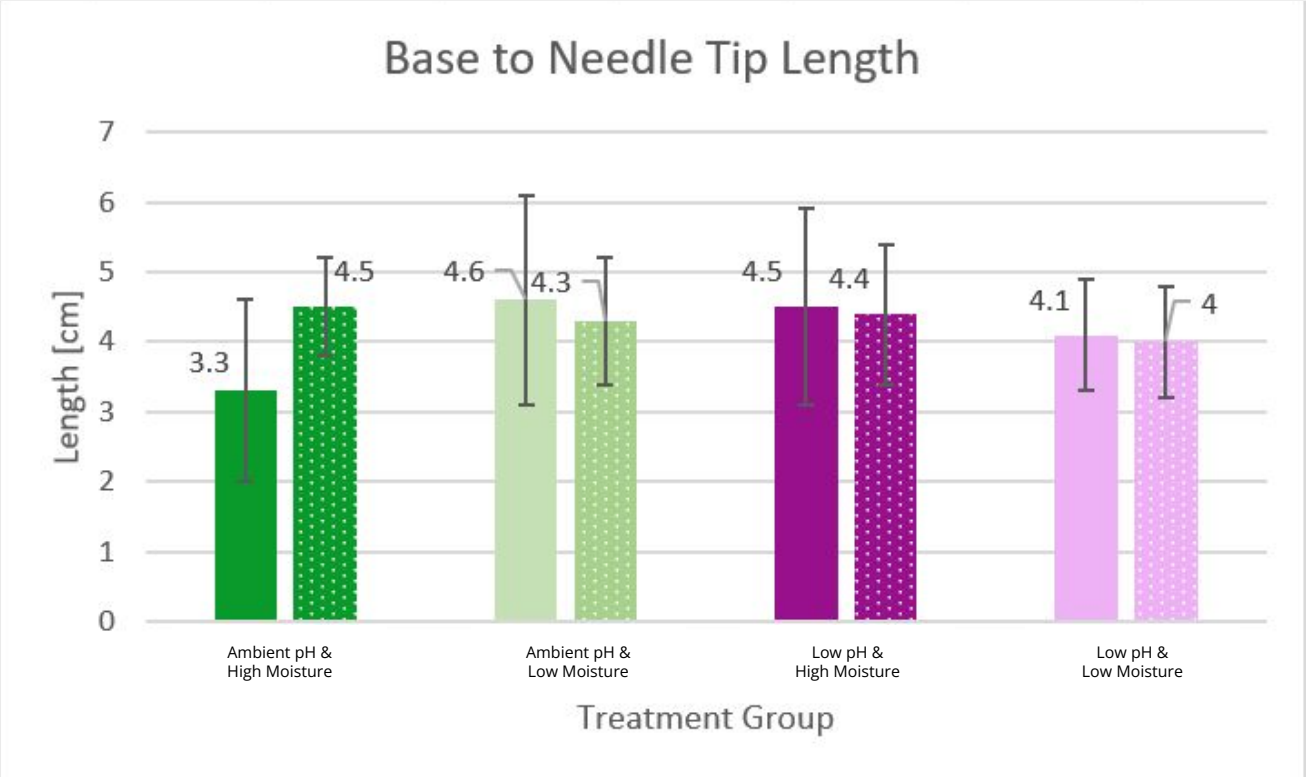


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=Symptomatic

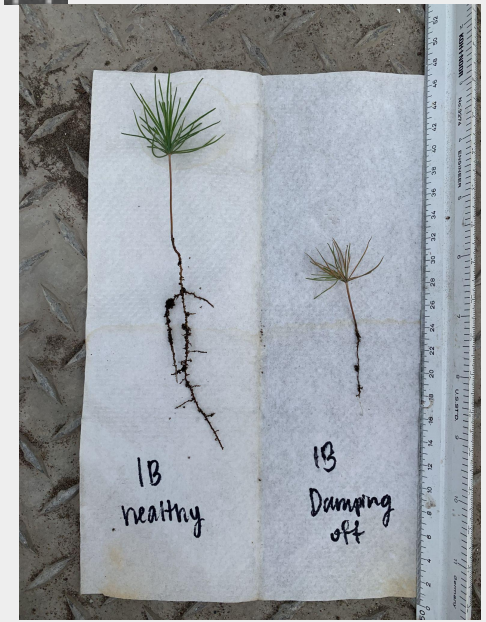


Results

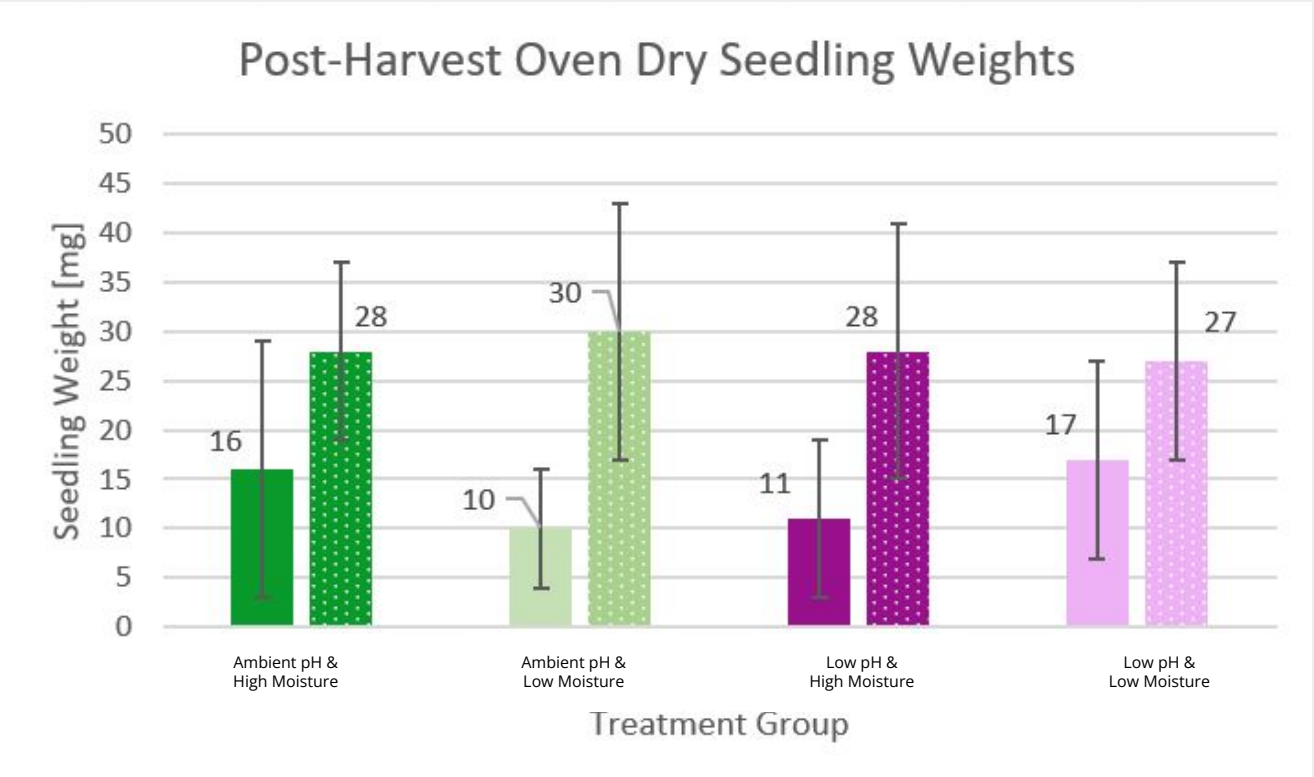


= Asymptomatic

=Symptomatic



Results



= Asymptomatic

=Symptomatic

Results

- Time between inoculation and seedling death
  - Bootstrap statistical test: p-values (H_0 = average day of seedling death is not different between populations) (alpha value of 0.05)
    - **Comparing pH:**
      - Ambient pH/High Moisture & Low pH/High Moisture: p value=0.5642
      - Ambient pH/Low Moisture & Low pH/Low Moisture: p value =0.9938
    - **Comparing moisture:**
      - Ambient pH/High Moisture & Ambient pH/Low Moisture: p value=0.1354
      - Low pH/High Moisture & Low pH/Low Moisture: p value=0.6632
  - Based on the bootstrap test, in each group the null hypothesis was not rejected, the timelines are not significantly different
- Number of seedlings that died in each group (population proportion test, alpha = 0.05)
  - Comparing Moisture:
    - **Ambient pH/High Moisture & Ambient pH/Low Moisture: p value= 0.035**
    - **Low pH/High Moisture & Low pH/Low Moisture: p value=0.018**
  - Comparing pH:
    - **Low pH/High Moisture & Ambient pH/High Moisture: p value= 0.008**
    - Low pH/Low Moisture & Ambient pH/Low Moisture: p value= 0.062 (accept H_0)

Discussion

- Proportion test
  - Moisture made a difference in both ambient and lower pH–
  - pH made a difference, but only with high moisture
    - **Overall: ambient pH and low moisture had the largest proportion of seedlings dead and the shortest time between inoculation and seedling death**
- Bootstrap test: no statistically significant difference in average time between death and inoculation
- Hypothesis was partially not supported
  - Frequency of death **increased as pH increased** (supported), but **decreased as soil moisture increased** (not supported)

Discussion

- Hypothesis partially supported
  - Most basic information suggests high moisture aids in excess fungal growth (Lamichhane et al., 2020; How to Prevent Seedling Damping Off, 2018)
  - In all categories, roots were found to be (on average) longer than the asymptomatic seedlings
    - Roots in infected seedlings should be almost nonexistent
    - The roots were still very thin compared to the asymptomatic roots, but even oven dried mass of symptomatic seedlings was higher in all treatment groups than asymptomatic seedlings
- Future studies should:
  - Have a bigger sample size– grow each seedling in a smaller container for more controlled environment
  - Use drip irrigation to ensure a steady supply of water instead of having to check moisture multiple times during the day
  - Each seedling in their own pot (more controlled environment)

Acknowledgement & References

Acknowledgements

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